Similarity

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

Similarity

> Prepare Similarity: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Calculate **similarity** using explicitly selected features and a stated scoring rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Confidence Is Not Correctness

Without opening the old lesson, explain one key idea from Confidence Is Not Correctness. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a confidence claim from **evidence** that a **prediction** was correct

Reference example, not the only acceptable wording: The learner replaced the documented score meaning with a probability interpretation. Report 80 of 100 feature-match points.

To claim a chance of correctness, a defined probability method and relevant outcome evaluation would be needed.

Boundaries and 'Not Sure'

Without opening the old lesson, explain one key idea from Boundaries and 'Not Sure'. Give an example and a mistake that the idea helps you avoid.

Look for: Identify a specific source of uncertainty and explain how it affects a **prediction**

Reference example, not the only acceptable wording: The rule and two stated possibilities are known; the actual height is unresolved. At 5 predict short, at 6 predict tall. Inspect the original readable record or remeasure.

Repeating a calculation supplies no new information about which height was recorded.

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a threshold by tracing **prediction**s and counting disagreements for every **candidate**

Reference example, not the only acceptable wording: Threshold 3 predicts short, tall, tall, tall, disagreeing only at 4 cm: count 1. Threshold 7 predicts short, short, short, tall, disagreeing only at 6 cm: count 1.

Reset the tally for each **candidate**.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A reader wants mystery and short books. A has both traits, B only mystery, C only short. Score one point per matching trait. The reader now says short is essential. Explain how the decision should change.

Concept to explain

Recommendations often begin by comparing selected traits, not whole objects.

Term	Meaning
similarity	degree of likeness under chosen traits
distance	numeric or ordered difference between representations
weight	importance assigned to a feature
nearest neighbor	item with smallest chosen distance

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Treat **similarity** as a defined computation with a purpose. This week uses exact category matching and whole-number **weight**s, avoiding formal **distance** geometry. Keep ranking, probability and eligibility distinct.

Students should be able to explain changes in results from the feature contributions rather than memorise a winning item.

Readiness answer

No. Read the score definition and selected features. Matching-feature points describe a comparison rule, not a probability of suitability.

Target and **candidate** card records from the lesson, with D and incomplete E added later

Counters for colour, shape and texture contributions

Book records J, K and L from W4 with feature headings

End goal for the pictorial case

By the end, I can calculate **similarity** under a declared feature rule and explain a changed ranking.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Similarity using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT represented traits for **candidate** items

2. Process — the deciding step

PROCESS compare **distance**s using selected features

3. Output — the result

OUTPUT nearest or most similar items

Key idea

CORE IDEA Recommendations often begin by comparing selected traits, not whole objects.

How the mechanism works

UNDER THE HOOD **Similarity** is a designed measure over represented features. **Distance** changes with feature selection, scaling, **weight**ing, and missing values. Nearest does not mean identical, best, or fair.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A book can be nearest to another by reading level, topic, length, or mood; the chosen features and scales determine the neighbor.

Step 1

STEP 1 — OBSERVE represented traits for **candidate** items

Step 2

STEP 2 — TRACE compare **distance**s using selected features

Step 3

STEP 3 — CONCLUDE nearest or most similar items

Model answer

MODEL REASONING Plot items consistently, define how each axis is measured, and compare **distance**s. A different axis or **weight** can change the nearest item without any item moving in the real world.

A claim to avoid

DO NOT OVERCLAIM **Similarity** is a tool designed for a purpose, not a universal property.

What new **evidence** could change this answer?

Concrete teaching case

The teaching case

Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

The teaching case

Rank by number of matching traits.

Exact explanation to model

Why the result follows

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

Explain the added worked example

Explain the added worked example · 1

Two objects can be alike in colour and different in shape. A computer comparison needs a precise answer to similar in what way. Select features, state how their values are compared, and define what a larger score means.

The result belongs to that rule; it is not a universal judgment about the objects.

Explain the added worked example · 2

Our target card is red, round and smooth. Three **candidate** cards are A: red, round, rough; B: blue, square, smooth; C: red, square, rough.

First give one point for each feature that exactly matches the target and zero for each known mismatch. Add the points. Colour, shape and texture are the only selected features.

This paper exercise uses complete category values with agreed spelling. Real records can be missing or uncertain. Do not silently treat an unknown texture as a known mismatch or copy a value from the target.

First identify whether the information supports the comparison.

A receives 1 for colour, 1 for shape and 0 for texture: total 2. B receives 0, 0 and 1: total 1. C receives 1, 0 and 0: total 1. Record the contributions, not just the totals, so another learner can check the computation.

A is most similar among these three **candidate**s under this rule. That sentence names the **candidate** set and the rule.

It does not say A is identical to the target, has the right texture or will be the best choice for a task that needs smooth material.

Key terms

Term	Meaning
Similarity	Closeness under a stated choice of features and comparison rule
Feature match	Agreement between two recorded values for a selected feature
Weight	The number of points assigned to a feature match in this scoring rule
Tie	Two or more candidates sharing the same score
Candidate	An item being considered in a comparison

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Candidate	Colour and shape points	Texture worth 3 points	New total
A	1 + 1	0	2

Candidate	Colour and shape points	Texture worth 3 points	New total
B	0 + 0	3	3
C	1 + 0	0	1

Model the reasoning aloud · 3

Suppose texture matters more for the stated comparison. Keep colour and shape worth one point each, but give three points for a texture match. B now scores 3 and becomes the unique highest scoring **candidate**.

No card changed; the rule changed. Record both versions rather than overwriting the first result.

Model the reasoning aloud · 4

Weights express priorities in this designed rule. They need a reason connected to the task and should be set before inspecting the final evaluation results. A convenient winner does not justify a **weight**.

If smooth texture is required without exception, use a requirement check rather than hoping a **weight**ed score will always enforce it.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Similarity

A fictional craft-card matcher compares each **candidate** with a target.

The facts we will use

Supplied facts

Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Supplied facts

Initially each exact **feature match** scores 1; return all highest scorers.

Supplied facts

For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

AI ACADEMY / AI-1/19

Declared weights change similarity

	All weights 1	Texture weight 3
A	2	4
B	2	4
C	2	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/19

Similarity

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Target features

Compare colour, shape and texture separately.



2. Equal weights

A matches colour+texture=2; B shape+texture=2; C colour+shape=2.



3. Initial result

All three tie at 2.



4. Texture weight 3

New scores A=4, B=4, C=2; A and B remain tied.

CHANGE ONE THING

Keep original weights but change the target texture to rough.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Target features	Compare colour, shape and texture separately.
2. Equal weights	A matches colour+texture=2; B shape+texture=2; C colour+shape=2.
3. Initial result	All three tie at 2.
4. Texture weight 3	New scores A=4, B=4, C=2; A and B remain tied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident.

What this example does not establish

Keep the stated limit

A high feature-match score does not establish usefulness or suitability beyond the features represented.

Change one thing in the pictured case

Change one thing

Change: Keep original **weight**s but change the target texture to rough.

Predicted result

Prediction: C scores 3; A and B each score 1.

Why the result follows

Reason: C now matches all three target features; the others lose their smooth-texture match.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Similarity, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Book A has animals and adventure; B has animals only; C has neither. A reader wants both traits.

2. State the decision to explain

Write the question to explain: Rank by number of matching traits.

3. Reveal the worked reasoning

Reveal after **prediction**s: A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

Hold back the independent task

Keep for an independent attempt: Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Original totals A 2, B 1, C 1, D 2, so return A and D. With texture worth 3: A 2, B 3, C 1, D 4, so return D. D combines shape and texture. Scores count **weight**ed matches, not observed chances of suitability.

Exit: the winner depends on **weight**s and **candidate** set; A's first result is not a universal fact.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve why a point score is not automatically a probability.
7 to 20 minutes	Trace the three diagram rows with one counter per match. State the target and candidates aloud.

Stage	Teacher action
20 to 33 minutes	Complete W1 and W2. Change only the texture weight and compare the two recorded results.
33 to 45 minutes	Use W3 to distinguish a missing value from a known mismatch. Require a recorded pause under the classroom policy.
Session 2 start 0 to 8 minutes	Retrieve the scoring rule and why it depends on selected features.
8 to 23 minutes	Introduce D and complete the guided task. Return all tied maximum scores and explain the weighted result.
23 to 38 minutes	Collect W4 independently. Look for all ties and the distinction between similarity and a required language.
38 to 45 minutes	Discuss W5 and the exit. End with the need to document feature choices and effective weights.

Reduce help deliberately

Completion practice

Completion task: Keep the same target and **candidate**s but make texture worth 3 points, with colour and shape worth 1 each. Recalculate A, B and C. Explain what changed and what stayed the same.

Reasoning to listen for

Teacher check: A stays at 2, B becomes 3 and C stays at 1, so B wins. Only the texture **weight** changed; **candidate** values, target and other **weight**s stayed the same. The result is specific to this new comparison rule.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Similarity** and selected features. Proposed explanation: “Similar has one objective meaning for every task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Name the features, comparison rule and purpose.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Point to the matches supporting C’s new score and explain why there is no tie.

Reasoned teacher answer

C matches red, round and rough: 3. A matches only red and B only round: 1 each.

Changed-case reasoning: C scores 3; A and B each score 1. C now matches all three target features; the others lose their smooth-texture match.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Similarity investigation and use the first unsupported step to choose a teaching response.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Create a trait map and find two different 'nearest' items.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

Find the unsupported claim

CLAIM TO REPAIR Two objects have one universal **similarity** score.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Similar has one objective meaning for every task: Name the features, comparison rule and purpose.

The highest score proves all requirements are met: Check mandatory requirements separately.

Equal scores can be resolved by whichever row appears first: Use the stated tie policy; here return all highest scorers.

Missing texture deserves zero as if it were rough: Retain an unknown and apply the stated missing-value policy.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Similarity questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

A scores 2, B 1, C 0. Ranking depends on the chosen traits; it is not a universal measure of book quality.

Independent question

Independent question · withhold the key

Compare books using topic only. Book A is about space with a red cover; B is about space with a blue cover; C is about dogs with a red cover. Which matches A best?

Expected independent answer

Teacher key · Expected independent answer

B matches A by topic. C shares cover colour, but colour was not the chosen comparison trait. **Similarity** depends on what we choose to compare.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Two objects have one universal **similarity** score.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Initial scores are A=2, B=1, C=1. With short essential, exclude B; A remains preferred to C if mystery still matters.

Similarity scores encode selected preferences; they do not establish universal quality or guarantee enjoyment.

Explanations for the additional workbook examples

W1 Show the contributions before the total

Workbook question · W1

For the red round smooth target and **candidate**s A, B and C in the lesson, record the colour, shape and texture contributions under one point per match. Give each total and identify every highest scoring **candidate**.

Teacher reasoning · W1

A: 1, 1, 0 gives 2. B: 0, 0, 1 gives 1. C: 1, 0, 0 gives 1. A is the only highest scoring **candidate** among A, B and C under equal **weight**s.

W2 Explain a changed winner

Workbook question · W2

Keep the same target and **candidate**s but make texture worth 3 points, with colour and shape worth 1 each. Recalculate A, B and C. Explain what changed and what stayed the same.

Teacher reasoning · W2

A stays at 2, B becomes 3 and C stays at 1, so B wins. Only the texture **weight** changed; **candidate** values, target and other **weight**s stayed the same. The result is specific to this new comparison rule.

W3 Keep unknown separate from different

Workbook question · W3

Candidate E is red and round but its texture is missing. A learner assigns zero for texture and declares E fully scored. Apply our classroom policy.

State the known contributions, the missing information and what must happen before a final ranking.

Teacher reasoning · W3

Colour and shape contribute 1 each, giving a known subtotal of 2 under the original rule. Texture is unknown, not a known mismatch.

Pause the final ranking, request E's texture, and record the unresolved feature if it cannot be obtained.

W4 Transfer the comparison to books

Workbook question · W4

A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish. Give 1 point per exact match and return all tied highest scorers.

Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners. Explain why a winner may still fail a requirement to read English.

Teacher reasoning · W4

Equal **weight**s: J 2, K 2, L 2; all three tie. Topic **weight**ed 3: J 4, K 2, L 4; J and L tie. L remains a highest scorer despite Spanish language, so **weight**ed **similarity** does not enforce the English requirement.

Check language eligibility separately if English is required.

W5 Notice an accidental extra weight

Workbook question · W5

A designer adds a second colour column that copies the first and gives both columns one point. They say no feature has been given extra importance because every column has equal **weight**.

Explain the problem and propose a clearer rule.

Teacher reasoning · W5

Colour now contributes twice for one underlying match, so it has effective **weight** 2 while other features have **weight** 1. Remove the duplicate or state and justify colour's **weight** explicitly.

Equal column **weight**s do not guarantee equal importance for the underlying features.

Assess independent transfer

Independent question · withhold the key

Selected case: A target book has topic space, format comic and language English. **Candidate**s are J: space, prose, English; K: animals, comic, English; L: space, comic, Spanish.

Give 1 point per exact match and return all tied highest scorers. Then make topic worth 3 points while keeping the other **weight**s at 1. Report both sets of totals and winners.

Explain why a winner may still fail a requirement to read English.

Selected case answer

Teacher key · Selected case answer

Equal **weight**s: J 2, K 2, L 2; all three tie. Topic **weight**ed 3: J 4, K 2, L 4; J and L tie. L remains a highest scorer despite Spanish language, so **weight**ed **similarity** does not enforce the English requirement.

Check language eligibility separately if English is required.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Similarity to Helpful Classroom Sorter, adapt support and plan a later changed-case check.

Helpful Classroom Sorter

Choose two useful item traits for **similarity**-based sorting and explain why each deserves its **weight**.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 20: Recommendations. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Place one row of counters under each feature heading. Keep the target visible while checking **candidate**s.

Extension: add a feature that is irrelevant to the stated purpose and show how it could change the winner without improving usefulness.

W4 is secure with 2/2/2 then 4/2/4, all **candidate**s then J and L as winners, and a separate English eligibility check. After support retry topic **weight** 2: totals 3/2/3, J and L still tie.

Require contribution explanations, not only totals.

Week 20 uses explicit scores and requirements to form recommendations, then examines what feedback can and cannot establish.

Plan the delayed return

Suggested return points in this level: ai-1/20, ai-1/22, ai-1/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Similarity lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Calculate **similarity** using explicitly selected features and a stated scoring rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Sections 1–6; record the brief recall in the workbook.

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>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Keep the same target and **candidate**s but make texture worth 3 points, with colour and shape worth 1 each. Recalculate A, B and C. Explain what changed and what stayed the same.

> ### Supported practice answer
>
> Reasoned practice answer: A stays at 2, B becomes 3 and C stays at 1, so B wins. Only the texture **weight** changed; **candidate** values, target and other **weight**s stayed the same. The result is specific to this new comparison rule.

Check every learner before discussion: Use the worked case for **Similarity** and selected features. Proposed explanation: “Similar has one objective meaning for every task”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Name the features, comparison rule and purpose.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Keep original **weight**s but change the target texture to rough. Require a **prediction** and a reason before comparing. Expected result: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

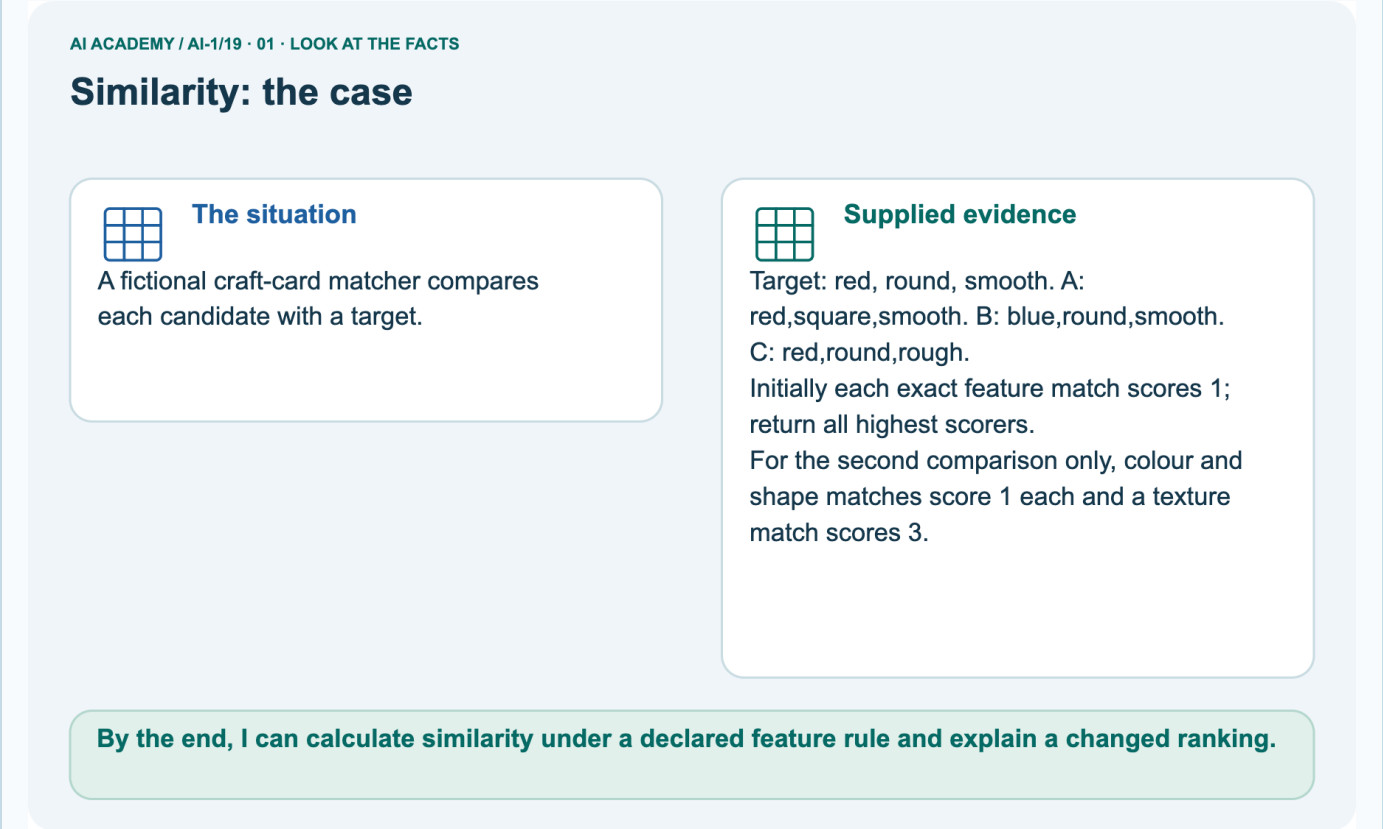
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Similarity graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Similarity: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft-card matcher compares each **candidate** with a target. Supplied facts: Target: red, round, smooth. A: red,square,smooth. B: blue,round,smooth. C: red,round,rough.

Initially each exact **feature match** scores 1; return all highest scorers. For the second comparison only, colour and shape matches score 1 each and a texture match scores 3.

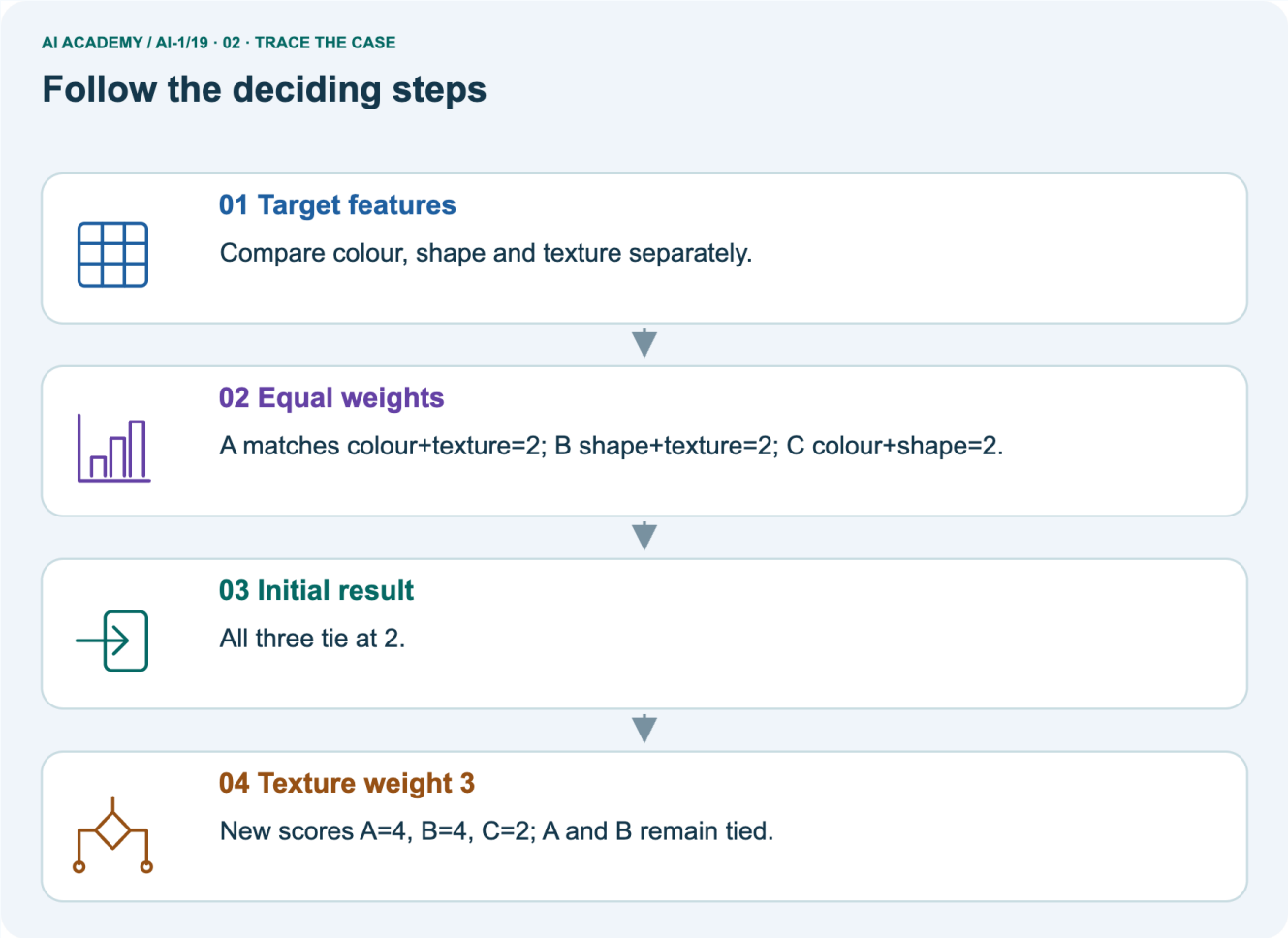
> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Target features: Compare colour, shape and texture separately. → Equal **weight**s: A matches colour+texture=2; B shape+texture=2; C colour+shape=2. → Initial result: All three tie at 2. → Texture **weight** 3: New scores A=4, B=4, C=2; A and B remain tied.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Declared weights change similarity



Follow the labelled picture with the matching explanation. Open the vector diagram

Declared **weight**s change **similarity**. A: All **weight**s 1 = 2, Texture **weight** 3 = 4. B: All **weight**s 1 = 2, Texture **weight** 3 = 4. C: All **weight**s 1 = 2, Texture **weight** 3 = 2.

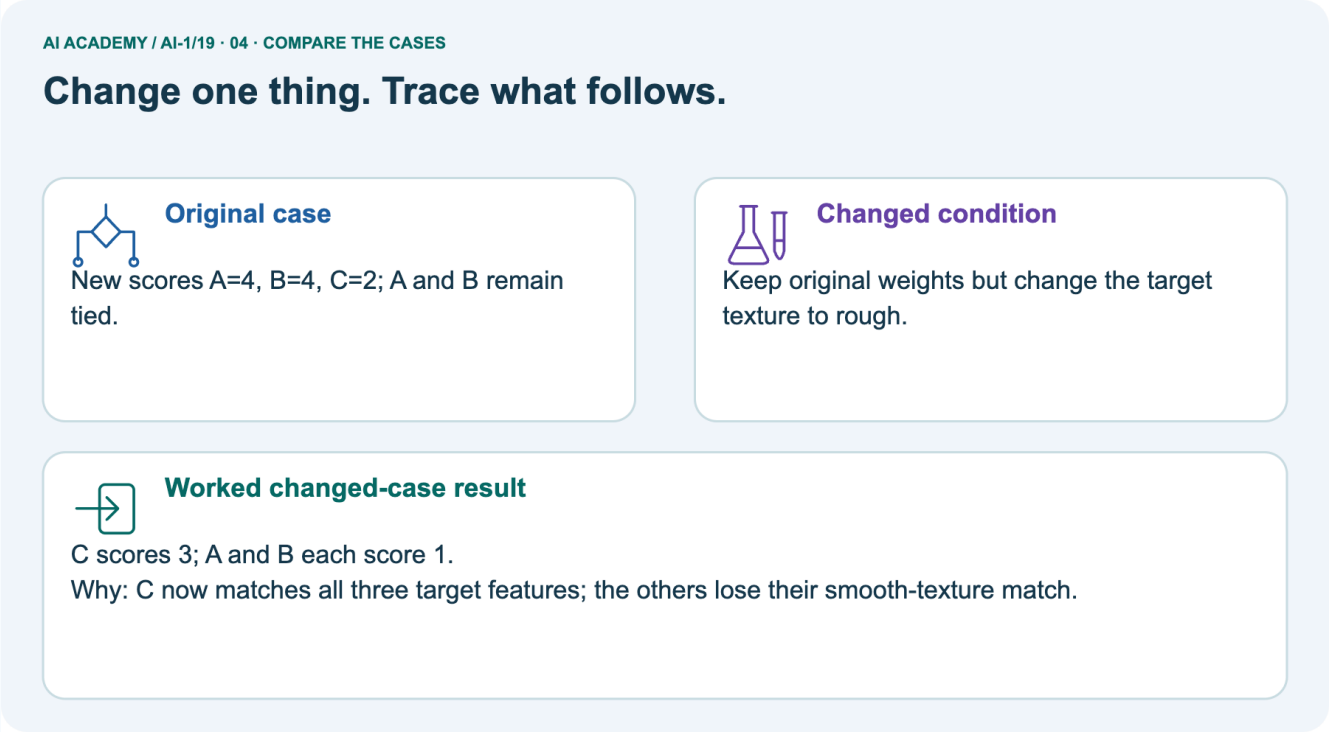
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: New scores A=4, B=4, C=2; A and B remain tied. Changed condition: Keep original **weight**s but change the target texture to rough. Expected result: C scores 3; A and B each score 1.

Reason: C now matches all three target features; the others lose their smooth-texture match.

Predict before checking the worked changed-case result. Name what changed and what the

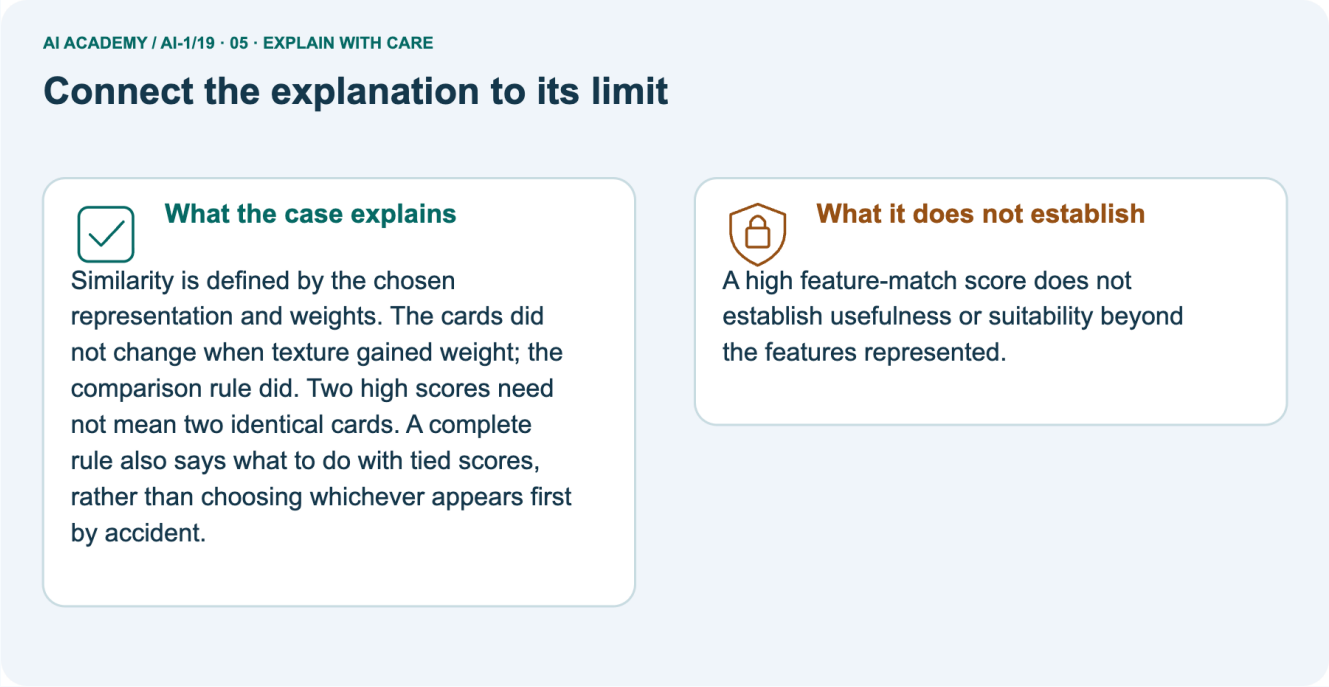
supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: C now matches all three target features; the others lose their smooth-texture match.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Similarity is defined by the chosen representation and **weight**s. The cards did not change when texture gained **weight**; the comparison rule did. Two high scores need not mean two identical cards.

A complete rule also says what to do with tied scores, rather than choosing whichever appears first by accident. Limit: A high feature-match score does not establish usefulness or suitability beyond the features represented.

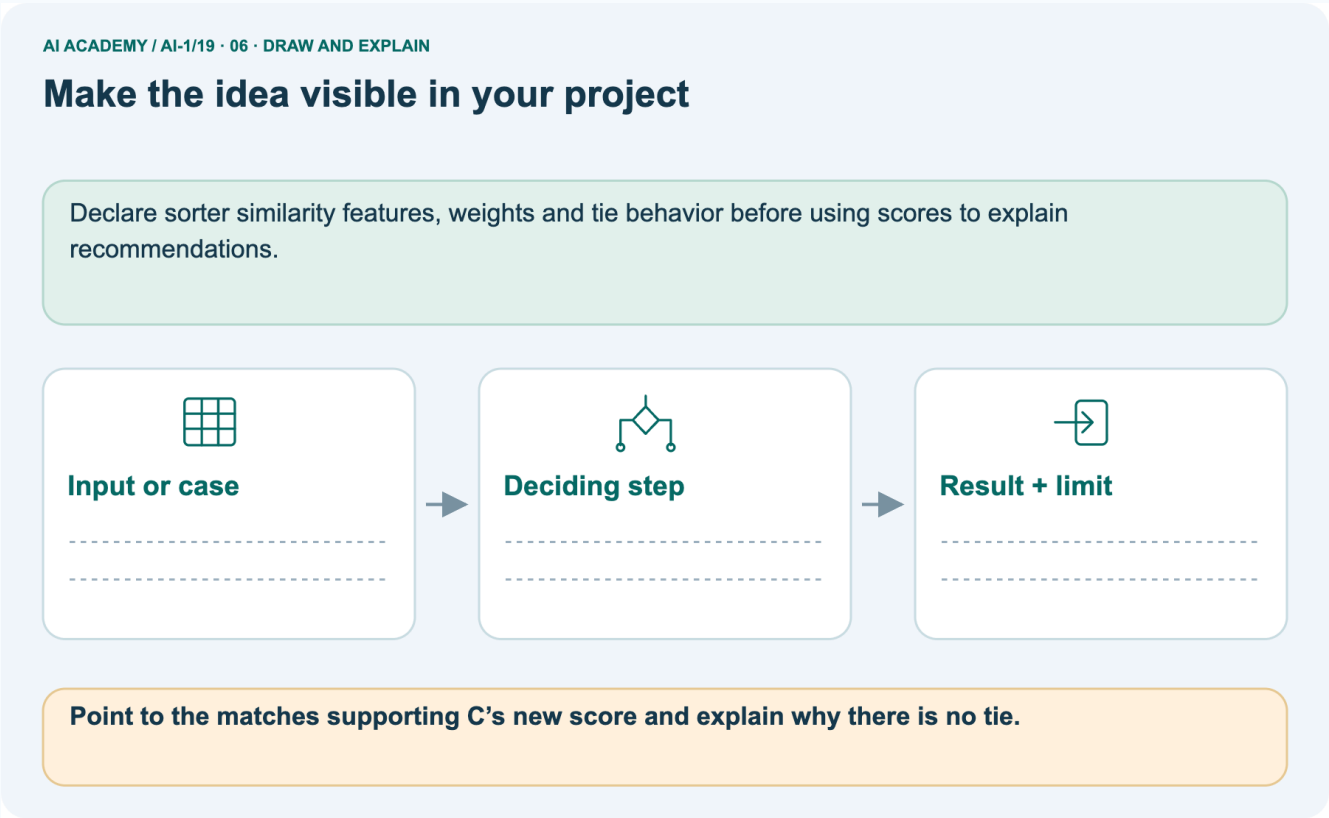
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A high feature-match score does not establish usefulness or suitability beyond the features represented.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare sorter **similarity** features, **weight**s and tie behavior before using scores to explain recommendations. Draw your **input**, deciding step, result and limit.

Supported question: Point to the matches supporting C's new score and explain why there is no tie.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Supported practice key: C matches red, round and rough: 3. A matches only red and B only round: 1 each.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.